CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 9



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Recap of Previous Lecture



- **Topic** Hardware Implementation of Mappings
- **Topic** Hit Latency
- **Topic** Mux & Comparators in Cache
- **Topic** Array Access with Cache

Topics to be Covered



Topic Array Access with Cache

Topic Multilevel Cache

[NAT]

#Q. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50 × 50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

How many data cache misses will occur in total?

32 blocks

Ans = 56

Array size = 2500 B

no. of blocks to store array $= \left\lceil \frac{2500\,B}{64\,B} \right\rceil = 40$

extra block = 40 − 32 = 8

	miss
first access	40
2nd access	2*8 = 16
	56



Ques) No. of hits in last Question?

Soln

2500 elements accessed twice.

Total mem. accesses = 5000

miss = 56

hits = 4944

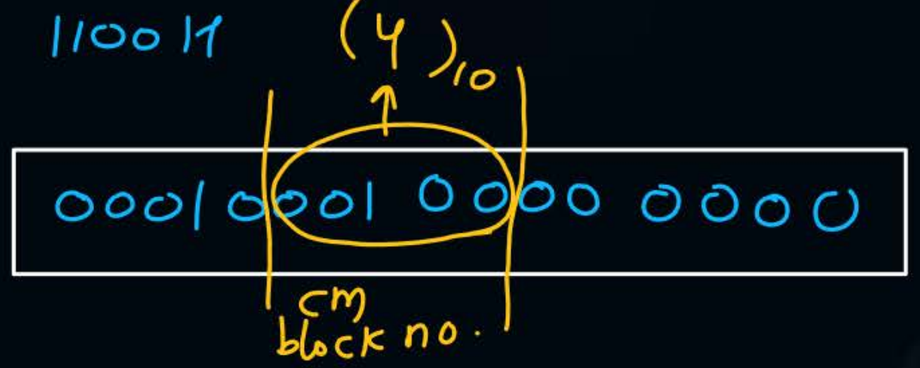
hit ratio = $\frac{4944}{5000}$

= 0.9888

= 98.88%.

[MCQ]

#Q. Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

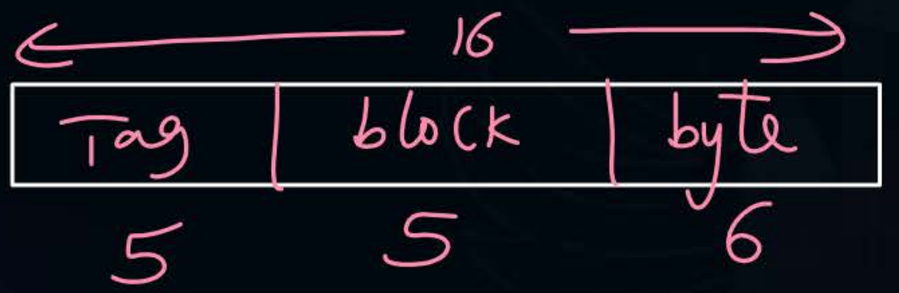


A ✓ line 4 to line 11

B line 4 to line 12

C line 0 to line 7

D line 0 to line 8



[NAT]

H.W.

#Q. Consider a direct mapped write back data cache of size 2KB with block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array A[30][25] with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 4 times. And between the accesses, there is no any data cache changes happen. Hit ratio of cache for the array access is?



Topic : Questions on Cache and Array

- Cache Size = 32 bytes
- Block size = 4 bytes

} no. of blocks in cache = $\frac{32B}{4B} = 8$

- Array in main memory A[8][8], each element is 1 bytes

• Direct mapping

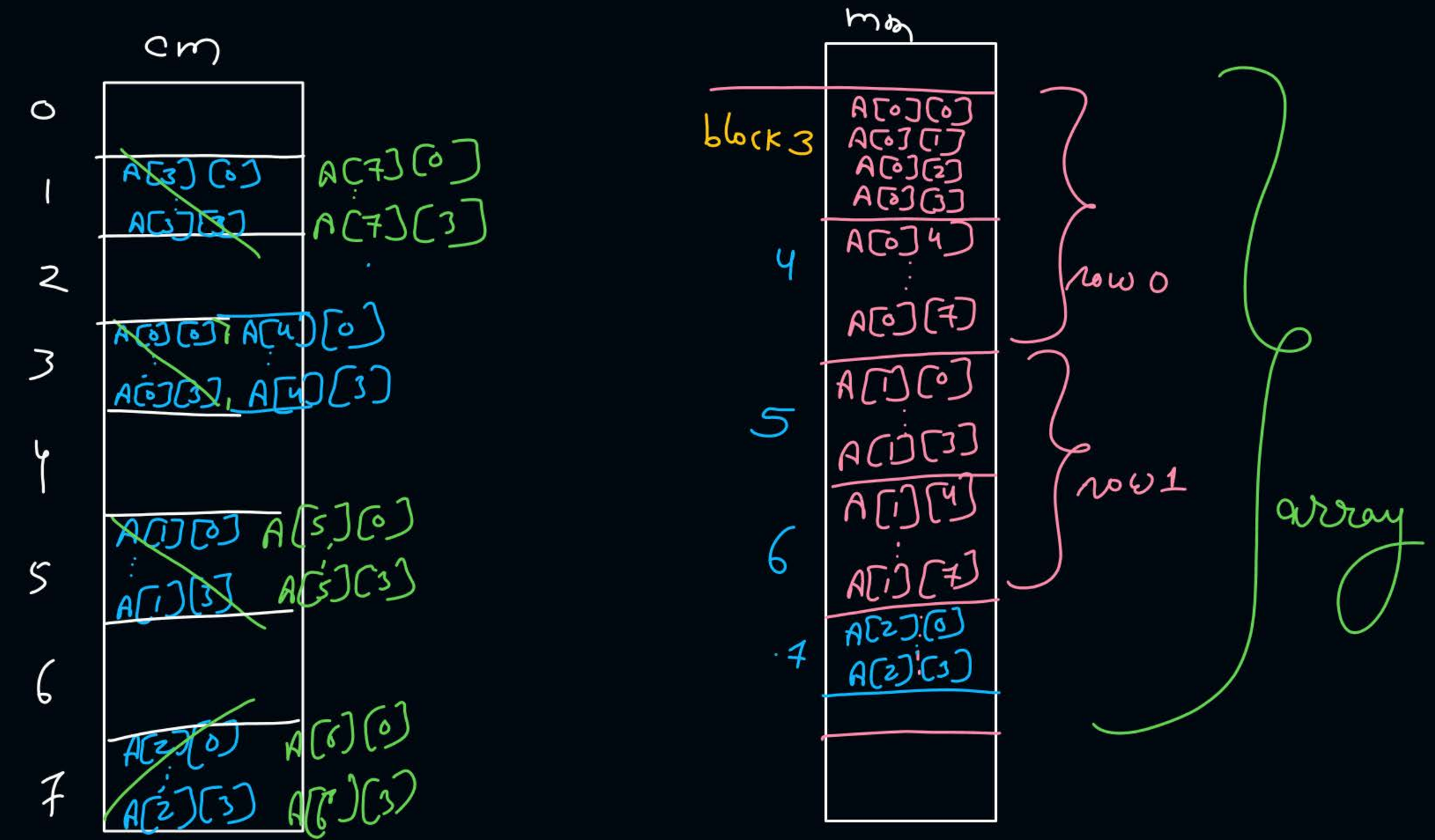
array size = 8 * 8 = 64 elements = 64B

no. of blocks for array = $\frac{64B}{4B} = 16$ Blocks

no. of elements in a block = $\frac{4B}{1B} = 4$

---

by default 2-d array stored in mem ⇒ Row-Major order


cm
0
1
2
3
4
5
6
7
A[3][0]
A[3][3]
A[7][0]
A[7][3]
A[0][0]
A[0][3]
A[4][0]
A[4][3]
A[1][0]
A[1][3]
A[5][0]
A[5][3]
A[2][0]
A[2][3]
A[6][0]
A[6][3]
mm
block 3
A[0][0]
A[0][1]
A[0][2]
A[0][3]
A[0][4]
A[0][7]
A[1][0]
A[1][3]
A[1][4]
A[1][7]
A[2][0]
A[2][3]
4
5
6
7
row 0
row 1
array

Access array column-wise :-

	Access	Hit/miss
Column 0	A[0][0]	miss
	A[1][0]	miss
	A[2][0]	miss
	A[3][0]	miss
	A[4][0]	miss
	A[5][0]	miss
	A[6][0]	miss
	A[7][0]	miss

all elements access will cause miss.

[NAT]

2^{15} B

2^{18} elements = $2^{18} * 8 = 2^{21}$ B

#Q. A CPU has a 32KB direct mapped cache with 128 byte-block size. Suppose A is two-dimensional array of size 512 × 512 with elements that occupy 8-bytes each. Consider the following two C code segments, P1 and P2.

P1 :

```
for (i = 0 ; i < 512 ; i ++)
{
    for ( j = 0 j < 512 j ++)
    {
        x += A[i][j] ;
    }
}
```

P2:

```
for ( i = 0 ; i < 512 i ++)
{
    for ( j = 0 j < 512 ; j ++)
    {
        x+= A[j] [i];
    }
}
```

no of blocks in cache = $\frac{2^{15} B}{128 B}$

$= 2^8$

$= 256$

no. of elements in a block $= \frac{128\,B}{8\,B} = 16$

no. of elements in a row $= 512$

no. of blocks needed to store one row $= \frac{512}{16} = 2^5 = 32$

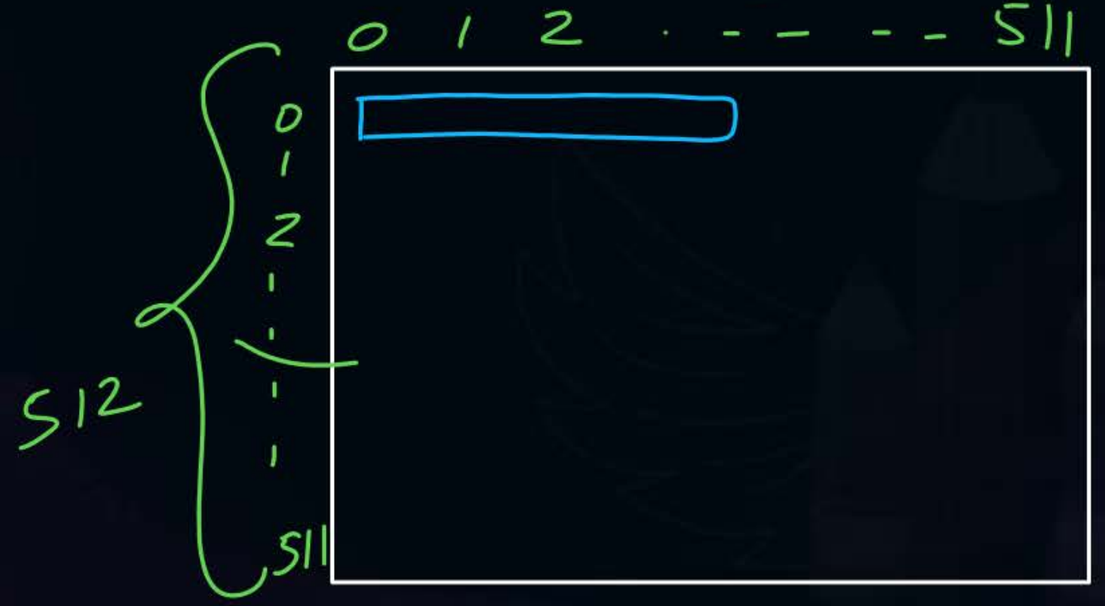
[NAT]

#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P_1 be M_1 and that for P_2 be M_2.

The value of M_1 is : = no. of blocks to store array

$$= \left\lceil \frac{2^{21} B}{128 B} \right\rceil$$

$$= 2^{14} \text{ Ans}$$



[NAT]

#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P_1 be M_1 and that for P_2 be M_2.

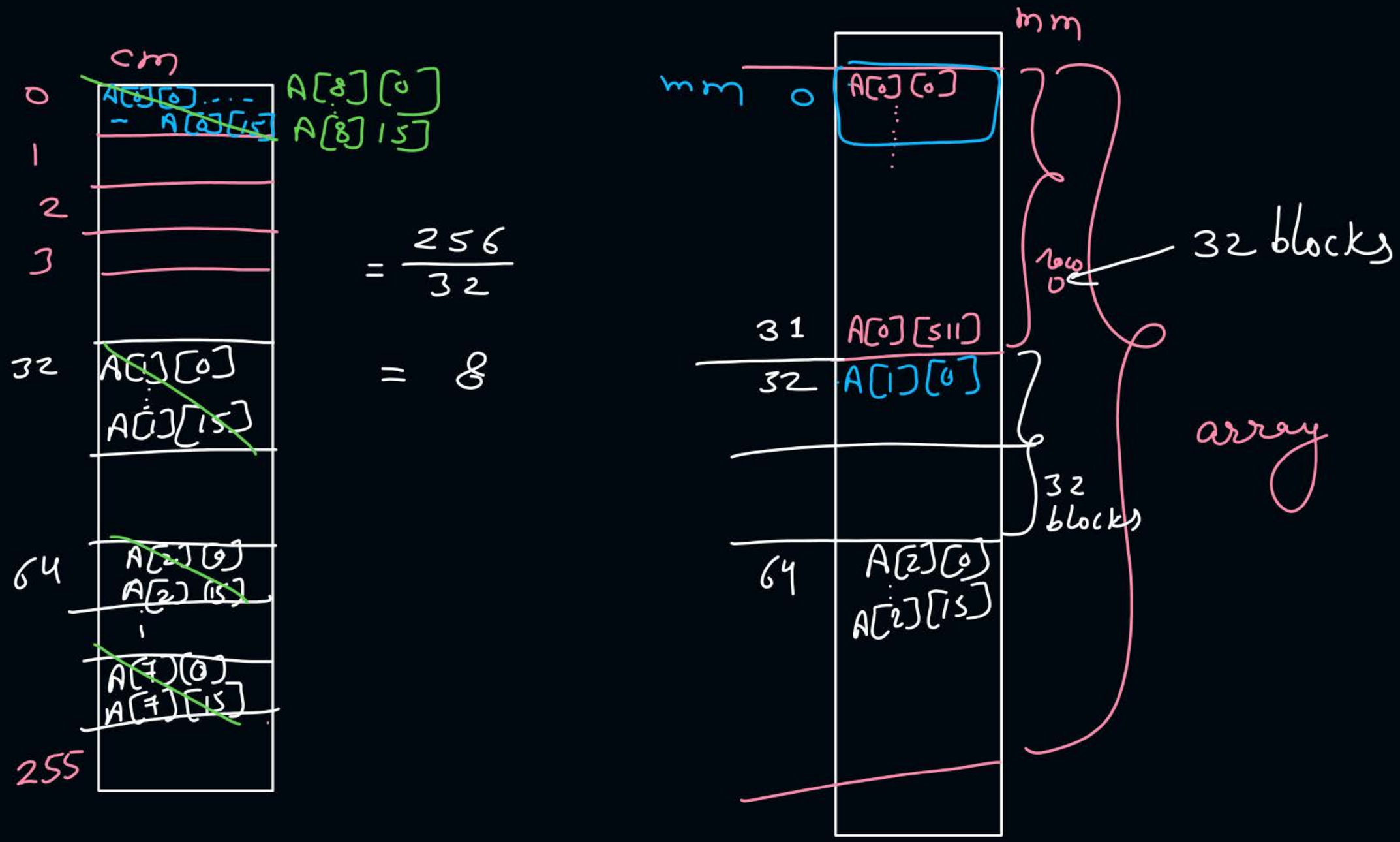
The value of M_2 is :

no. of miss = no. of elements in array

$= 2^{18}$

hit ratio = 0 %

miss ratio = 100 %


cm
0
A[0][0] ... -
A[0][15]
A[8][0]
A[8]15]
1
2
3
= 256/32
32
A[1][0]
A[1][15]
= 8
64
A[2][0]
A[2][15]
A[7][0]
A[7][15]
255
mm
mm
0
A[0][0]
31
A[0][511]
32
A[1][0]
32 blocks
1040
0
32 blocks
array
64
A[2][0]
A[2][15]

Access	Hit/miss
A[0][0]	miss
A[1][0]	miss
A[2][0]	miss

[MSQ] 14.03. [2022 : 2M]

#Q. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

A Every access to S is a hit.

B Once P is brought to the cache it is never evicted.

C At the end of the execution only R and S reside in the cache.

D Every access to R evicts Q from the cache.



Topic : Goal of Using Cache

1. Minimize Access Time → Use smaller cache
2. Maximize Hit Rate → Use larger cache

} multilevel cache

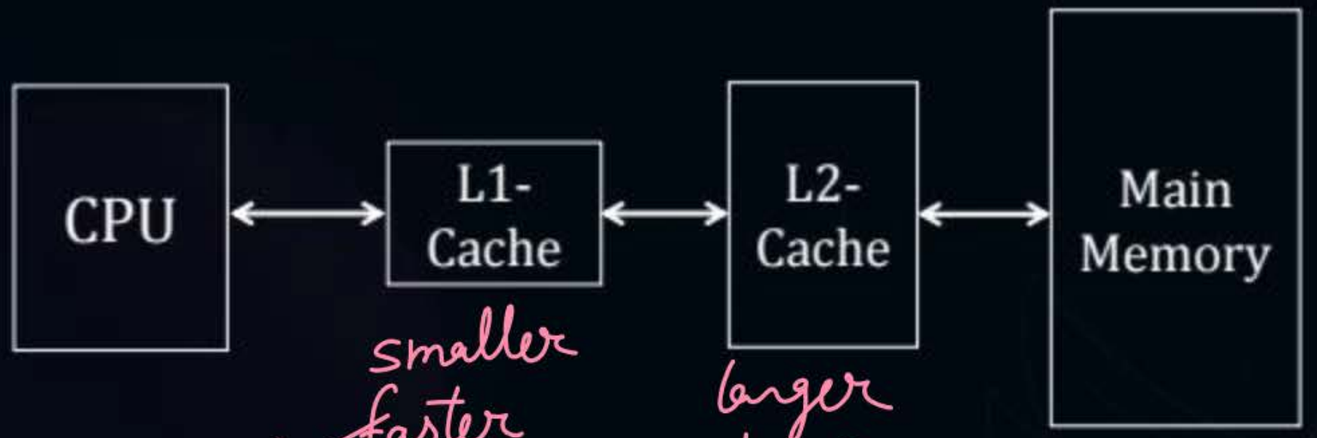
3. Minimize Miss Penalty



Topic : Multilevel Cache

Hit H_1 H_2

Access time t_1 t_2 t_{mm}



smaller faster low hit rate

larger slower high hit rate



Topic : Average Access Time Multilevel Cache

Sim.

$$t_{avg} = H_1 * t_1 + (1-H_1)\left[H_2 * t_2 + (1-H_2)\, t_{mm}\right]$$

or

$$= H_1 t_1 + (1-H_1) H_2 * t_2 + (1-H_1)(1-H_2)\, t_{mm}$$

Hier. :-

⇓

(default)

$$t_{avg} = H_1 t_1 + (1-H_1)\left[H_2 (t_1+t_2) + (1-H_2)(t_1+t_2+t_{mm})\right]$$

or

$$= H_1 t_1 + (1-H_1) H_2 (t_1+t_2) + (1-H_1)(1-H_2)(t_1+t_2+t_{mm})$$

or

$$= t_1 + (1-H_1)\left[t_2 + (1-H_2)\, t_{mm}\right]$$

[NAT]

#Q. Consider a 3-level memory hierarchy with L1 cache, L2 cache and a main memory. The hit ratios of L1 is 90% and of L2 is 95%. The access times of L1, L2 and main memory are 15ns, 60ns and 350ns respectively. The average memory access time is _____ns?

$$T_{avg} = 15 + 0.1 * [60 + 0.05 * 350]$$

$$= 22.75 \text{ ns}$$

Ques)
In prev. Questⁿ if only L1 and mm used



$$t_{avg} = 0.9 * 15 + 0.1 [15 + 350]$$
$$= 15 + 35$$
$$= 50 \text{ ns}$$

Ques) In prev questⁿ if only L2 and mm used



$$t_{avg} = 60 + 0.05 * 350 \text{ ns}$$
$$= 77.5 \text{ ns}$$



Topic : Probability of Access

fraction of access fulfilled from L1 = h_1

" " " L2 = $(1-h_1) * h_2$

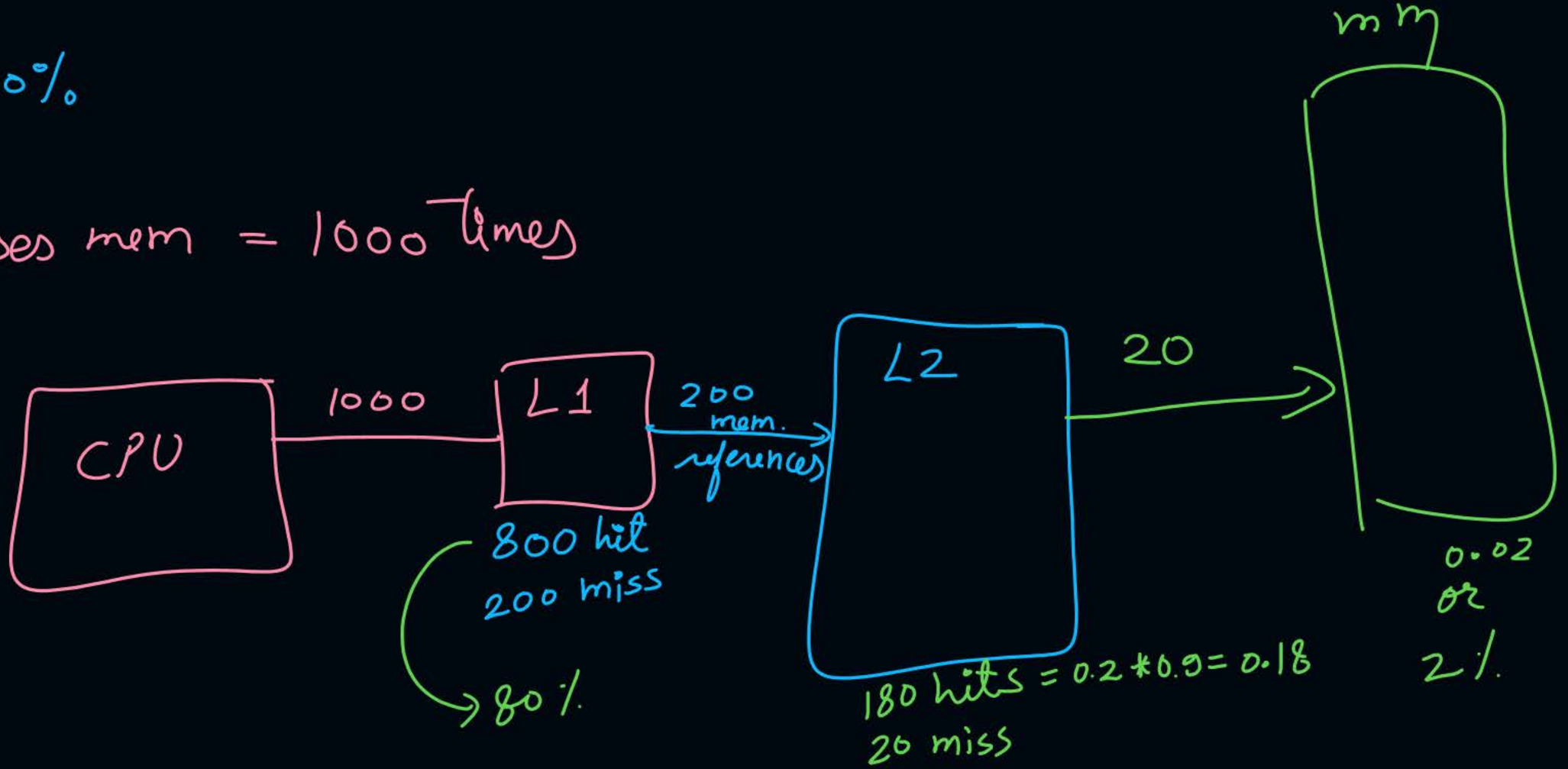
" " " " mm = $(1-h_1) * (1-h_2)$

eg:-

$h_1 = 80\%$

$h_2 = 90\%$

CP accesses mem = 1000 times



[NAT]

#Q. Consider a 3-level memory hierarchy with L1 cache, L2 cache and a main memory. The probability of access of L1 is 95%, of L2 is 4.5% and of main memory is 0.5%. The access times of L1, L2 and main memory are 10ns, 50ns and 400ns respectively. The average memory access time is _____ns?

Calculate for simultaneous and hierarchical both.

sim:- $t_{avg} = (0.95 * 10) + (0.045 * 50) + (0.005 * 400)$

$= 13.75$ ns

---

hier:- $t_{avg} = (0.95 * 10) + 0.045 * (10 + 50) + 0.005 * (10 + 50 + 400)$

$= 14.5$ ns



2 mins Summary

Topic Array Access with Cache

Topic Multilevel Cache



Happy Learning

THANK - YOU